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ORIGINAL ARTICLE

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Tumor-specific immunity in MUC1.Tg mice induced by immunization with peptide vaccines from the cytoplasmic tail of CD227 (MUC1)

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Abstract Purpose: CD227 (MUC1), a membrane-associated glycoprotein expressed by many types of ductal epithelia, including pancreas, breast, lung, and gastrointestinal tract, is overexpressed and aberrantly glycosylated by malignant cells. We sought to define epitopes on MUC1 recognized by the different cell-mediated immune responses by an in vivo assay. Epitopes identified by this assay were evaluated for efficacy to protect mice transgenic for human MUC1 (MUC1.Tg) against MUC1-expressing tumor growth. Methods: We investigated contributions of the tandem repeat (TR) and the cytoplasmic tail (CT) of MUC1 to the MUC1-specific immunological rejection of tumor cells. MUC1 cDNA constructs, in which the TR region was deleted or the CT was truncated, were transfected into two different murine tumor cell lines (B16 and Panc02), which were used to challenge mice and evaluate immunological rejection of the tumors. We used tumor rejection in vivo to define epitopes on the TR and CT of MUC1 recognized by T cell–mediated immune responses in a preclinical murine model. Results: Our findings demonstrated that the TR and a portion of the MUC1 CT contributed to $CD4^+$ T cell rejection of MUC1-expressing B16 tumor cells, but not rejection of MUC1-expressing Panc02 tumor cells. A separate epitope in the CT of MUC1 was necessary for $CD8^+$ T cell rejection of Panc02 tumor cells. Based on these studies, we sought to evaluate the efficacy of immunizing mice transgenic for (and immunologically tolerant to) human MUC1 with peptides derived from the amino acid sequence of the CT of MUC1. Results showed that survival can be significantly prolonged in vaccinated MUC1.Tg mice challenged with MUC1-expressing tumor cells, without induction of autoimmune responses. Conclusions: These studies demonstrated that MUC1 peptides may be utilized as an effective anticancer immunotherapeutic, and confirmed the importance of immunogenic epitopes outside of the TR.

Keywords Antitumor immunotherapy · In vivo epitope mapping · MUC1 · Tumor-associated antigens · Tumor immunity

Abbreviations aa: Amino acid · CT: Cytoplasmic tail · IFN-γ: Interferon gamma · MUC1.Tg: MUC1 transgenic mice · TR: Tandem repeat · wt: Wild-type C57BL/6 mice

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Introduction

MUC1 is a large, transmembrane glycoprotein expressed on the apical surface of many types of polarized epithelia including pancreas, lung, breast, and the gastrointestinal tract [1]. MUC1 is overexpressed and differentially glycosylated by a number of adenocarcinomas [2–5]. Humoral and cell-mediated immune responses against MUC1 are detected in patients with $MUC1^+$ tumors, as measured in vitro [6–8]; however, these responses are often ineffective at eliminating the tumors in vivo.

A number of MUC1-based immunogens have been evaluated as potential cancer vaccines [9–12], including cells, purified antigen, and peptide or glycopeptide fragments derived from the tandem repeat (TR) region [1, 13]. Clinical trials of vaccines have focused on the TR region [1, 10, 11, 13]. Recently, putative epitopes from

regions outside of the TR of MUC1 have also been investigated [14–16], including the signal sequence and degenerate TRs. However, other potentially important epitopes from this tumor-associated antigen, especially those in the cytoplasmic tail (CT), have not been extensively studied. The endpoints of most studies have been in vitro assays, which determined that the TR region contained immunodominant epitopes for production of MUC1-specific antibodies and cytotoxic T lymphocytes (CTLs). Studies in our laboratory showed that current in vitro assays of cytolytic responses do not accurately predict MUC1-specific tumor rejection [17]. For example, no detectable differences were observed in the anti-MUC1 CTL precursor frequencies of wild-type (wt) C57BL/6 mice and C57BL/6 mice transgenic for human MUC1 (MUC1.Tg) [17], although wt mice rejected MUC1-expressing tumors in a MUC1-specific manner, while MUC1.Tg mice did not reject these tumors and showed evidence of immunological tolerance to MUC1 [17, 18].

The nature of cellular immune responses in vivo that mediate rejection of MUC1-expressing tumors in mice was previously investigated by experiments that depleted $CD4^+$, $CD8^+$, or both T cell subsets in vivo. $CD4^+$ cells were required for elimination of a human MUC1-expressing murine melanoma cell line (B16.MUC1), and $CD8^+$ cells were required for the elimination of a human MUC1-expressing murine pancreatic carcinoma cell line (Panc02.MUC1), in wt C57BL/6 mice [19, 20]. Using mice deficient in molecular components critical to the immune responses, further studies confirmed and extended these observations [21, 22] by showing that both $CD4^+$ and $CD8^+$ responses were mediated by α/β T cell receptors and required costimulation through CD28, as well as interactions between CD40 and CD40 ligand, and the activities of interferon γ (IFN-γ), and lymphotoxin α. A number of other factors (IL-4, IL-10, IL-12, and TNFR-1) were not required. There were differences in the effector mechanisms as the CD8-mediated cytotoxicity required perforin but not FasL; in contrast, the CD4-mediated cytotoxic response required both perforin and FasL.

In studies described herein, we sought to identify epitopes of MUC1 that were recognized and required by the MUC1-specific $CD4^+$ and $CD8^+$ T cell populations. Putative epitopes were mapped by challenging mice with tumor cell lines (B16 and Panc02) that expressed constructs of human MUC1 in which portions of the coding sequences for the protein were deleted. The C-terminus of the CT and the TR portion of MUC1 were required for rejection of B16.MUC1, while rejection of Panc02.-MUC1 required a distinct portion of the CT of MUC1, and not the TR. Based on results obtained from these studies, we hypothesized that vaccination with peptides derived from the amino acid (aa) sequence of MUC1 CT would generate protective immune responses against MUC1-expressing tumors in MUC1.Tg mice. Survival of MUC1.Tg mice challenged with MUC1-expressing B16 or Panc02 tumor cells was significantly prolonged following vaccination with three overlapping peptides spanning the entire CT portion of MUC1. Furthermore, vaccination with MUC1 CT peptides did not induce detectable autoimmune responses. These results demonstrate the importance of immunogenic epitopes outside of the TR and suggest that immunization with MUC1 CT peptides is an effective anticancer immunotherapeutic strategy.

Materials and methods

Cell culture

The BL6 variant of the C57BL/6 derived murine melanoma cell line B16 was maintained in Dulbecco's modified Eagle's medium (DMEM, Gibco; BRL, Invitrogen, Rockville, MD, USA) supplemented with 10% heat-inactivated fetal bovine serum (BioWhittaker, Walkersville, MD, USA), essential aa (BioWhittaker), non-essential aa (Biowhittaker), sodium pyruvate (Sigma, St Louis, MO, USA), vitamins (Gibco), penicillin/streptomycin (BioWhittaker). Panc02, a methylcholanthrene-induced pancreatic carcinoma syngeneic to C57BL/6, was a generous gift from Dr J. Nelson (University of Texas M.D. Anderson Cancer Center, Houston, TX, USA), and maintained in McCoy's 5A medium (Gibco) supplemented with 10% heat-inactivated fetal bovine serum, and penicillin/streptomycin in a humidified incubator at 37°C and 5% CO_2. EL4 cells were cultured in RPMI 1640 medium (Gibco) supplemented with 10% heat-inactivated fetal bovine serum and penicillin/streptomycin in a humidified incubator at 37°C and 5% CO_2. Culture media for MUC1 transfectant clones of B16, Panc02, and EL4 were supplemented with 600 μg/ml G418 (Mediatech, Herndon, VA, USA).

Expression of epitope-tagged MUC1 deletion constructs

B16 and Panc02 were transfected with plasmid DNA encoding a full-length human MUC1 cDNA (B16.MUC1 or Panc02.MUC1) or control expression vector (B16.neo, or Panc02.neo) as described previously [18, 23]. The full-length CT is 69 aa in length. MUC1 CT deletion constructs were generated as described previously [24, 25]. The TR comprises a 20 aa sequence (PDTRPAPGSTAPPAHGVTSA) that repeats 42 times. The MUC1 TR deleted (ΔTR) construct was generated as described previously [24, 25]. Flag epitope-tagged human MUC1 cDNA constructs in which portions of the CT (MUC1.CT3, MUC1.CT33, MUC1.CT45), or TR (MUC1[ΔTR]) of MUC1 were deleted and subcloned into the expression vector pHβ-Apr1-neo were described previously [24–26]. B16 and Panc02 were transfected with plasmid DNA using the Lipofectin method (Gibco; BRL, Rockville, MD, USA) or the GenePorter 2 method (Gene Therapy Systems, San

Diego, CA, USA). Cells were plated in 100 mm tissue culture dishes (Falcon Plastics, BD Labware, Franklin Lakes, NJ, USA) and grown to approximately 70% confluence. For the Lipofectin method, growth medium was removed, and the cells were washed with 1×PBS, and incubated 12 h in 5 ml of serum-free DMEM with 10 μl Lipofectin reagent and 10 μg plasmid DNA linearized by *Sca*I digestion. After 48 h, the transfection medium was replaced with selection medium containing 600 μg/ml G418. For the GenePorter 2 method, growth medium was removed, and the cells were washed with 1×PBS, and then incubated 48 h in 5 ml of serum-free DMEM containing 42 μl GenePorter 2 reagent, 12 μg linearized plasmid DNA linearized by *Sca*I digestion, and 300-μl DNA diluent B (Gene Therapy Systems). Transfection reactions of Panc02 cells using GenePorter 2 also contained 50 μl transfection booster 3 (Gene Therapy Systems). After 48 h, the transfection medium was replaced with selection medium containing 600 μg/ml G418. After approximately 7–10 days, single colonies were selected with cloning cylinders and expanded. Clonal cell lines were evaluated for coexpression of Flag epitope and MUC1 epitopes by western blotting with anti-Flag antibody (M2) (Sigma, St Louis, MO, USA) and anti-MUC1 antibodies HMFG-2 or CT-2 (both gifts from Dr Sandra Gendler, Mayo Clinic, Scottsdale, AZ, USA). Cell lines found to express the Flag epitope and MUC1 epitopes by Western blotting, were evaluated for Flag epitope and MUC1 epitope surface expression by flow cytometry.

Preparation of cell lysates

Cell lysates were prepared by scraping cells into 1 ml of lysis buffer (10 mM Tris, 150 mM NaCl, 1 mM phenylmethylsulfonyl fluoride, 1% Triton X-100) with a rubber cell scraper. Lysates were incubated on ice for 30 min and centrifuged at 4°C for 2 min at 6,000 rpm to remove cell debris. Supernatants were transferred to fresh tubes, and protein content was determined using the Bio-Rad protein assay (Bio-Rad, Hercules, CA, USA) with bovine serum albumin standards. Cell lysates were stored at −20°C.

Immunoblotting

Cell lysates were resolved on 10% denaturing polyacrylamide gels (with 3% polyacrylamide stacking gel), transferred to polyvinylpyrrolidine difluoride membranes electrophoretically, and blocked overnight in blotto (5% dry milk in 1×TBS [0.9% NaCl, 10 mM Tris, pH 7.4, 0.5% $MgCl_2$]). Primary antibodies were diluted 1:2,000 in blotto. Incubations were for 1 h at room temperature with light shaking, followed by three 10-min washes with blotto. Alkaline phosphatase–conjugated goat antimouse secondary antibodies were diluted 1:2,500 in blotto, and incubations were for 1 h at room temperature with light shaking. Following incubation with secondary antibody, the membranes were washed three times as above. Enhanced chemifluorescence (ECF) reagents were applied as per manufacturer's instructions (Amersham Life Science, Buckinghamshire, UK), and blots were visualized using a Fujifilm LAS-1000 CCD imaging system (Fuji Film, Tokyo, Japan). Analysis was performed using IR-LAS-1000 Lite V.1.1 software (Fuji).

Flow cytometry

Flow cytometric analysis of MUC1 surface expression on B16 and Panc02 transfectants was performed as follows. Adherent cells were released from tissue culture flasks by treating with 0.05 mM trypsin and 1.5 mM EDTA in PBS for 5 min at 37°C. All subsequent steps were carried out on ice. The cells were resuspended in FACS medium (1×PBS, 0.2% BSA, 0.1% sodium azide) at a concentration of 1×10^6 cells/ml, and incubated with M2 antibody or anti-MUC1 TR antibody (HMFG-2) for 20 min at 4°C. The cells were washed with FACS medium and incubated with a phycoerythrin-conjugated (PE) rabbit antimouse secondary antibody (Jackson ImmunoResearch Laboratories, West Grove, PA, USA) for 30 min at 4°C. The cells were washed again and resuspended in FACS medium, followed by analysis on a FACSCalibur (Becton Dickinson, Mountain View, CA, USA). Analysis was performed with Cell Quest software (Becton Dickinson).

Mice

Male and female wt C57BL/6 mice were purchased from the National Cancer Institute (Frederick, MD, USA). Age-matched MUC1.Tg mice were obtained from breeding colonies at the University of Nebraska Medical Center (Omaha, NE, USA). Mice were treated in accordance with Institutional Animal Care and Use Committee guidelines.

Tumor challenge

On the day of tumor challenge, adherent control and MUC1-expressing B16 tumor cell lines were released from tissue culture flasks by treating with 0.05 mM trypsin and 1.5 mM EDTA in PBS for 5 min at 37°C, counted, and resuspended in DMEM at a concentration of 2×10^5 viable cells/ml. Control and MUC1-expressing Panc02 tumor cell lines were similarly prepared except that they were resuspended at 1×10^7 viable cells/ml. B16.MUC1, B16.MUC1.CT3, B16.MUC1.CT33, B16.MUC1.CT45, B16.MUC1(ΔTR), or B16.neo 2×10^4 viable cells were injected subcutaneously (s.c.) between the scapulae. Panc02.MUC1, Panc02.MUC1.CT3, Panc02.MUC1.CT33, Panc02.MUC1.CT45, Panc02.-MUC1(ΔTR), or Panc02.neo 1×10^6 viable cells were

injected s.c. between the scapulae. Tumor growth was evaluated every 2–3 days, and tumor diameter was measured using a caliper. Kaplan-Meier survival curves are shown for the tumor challenge studies. Death was defined as the date on which the tumor diameter measured 10 mm. Mice were euthanized when the tumor diameter exceeded 10 mm. The log-rank test and Cox regression analyses were used for statistical analyses. The greatest (and most difficult to control) variable in these experiments is slight variation in the number of injected viable tumor cells (due to uncontrolled factors in cell processing on different days) that would alter the survival time. To minimize this variable, our experimental design was to always initiate experiments shown in the figures, at the same time points so that vaccinated and control groups were challenged with the same tumor cell preparation at the initiation of each experiment, and to compare survival times for experimental and control groups initiated at the same time.

Prediction of potential epitopes on the amino acid sequence of MUC1 CT, using two Web-based algorithms

We used two Web-based algorithms to analyze the aa sequence of MUC1 CT for potential human and murine MHC class I– and class II–binding epitopes. The first algorithm, SYFPEITHI, was developed by Rammensee et al. [27] and is available at http://www.uni-tuebingen.de/uni/kxi/. This algorithm ranks peptides according to a score taking into account the presence of primary and secondary MHC-binding anchor residues. The second algorithm, BIMAS, was developed by Parker et al. [28] and is available at http://bimas.dcrt.nih.gov/molbio/hla_bind/. This algorithm ranks potential binding according to the predicted half-time of dissociation of peptide/MHC complexes. Both algorithms are in the public domain and accessible through the Internet.

Synthetic peptides

The peptides used in this study were CQCRRKNYGQLDIFPARDTYHPMSEYPTYH, a 30 aa peptide corresponding to MUC1 CT aa residues 3–27; HPMSEYPTYHTHGRYVPPSSTDRSPYEKVSAG, a 32 aa peptide corresponding to MUC1 CT amino acid residues 18–49; STDRSPYEKVSAGNGG SSLSYTNPAVAAASANL, a 33 aa peptide corresponding to MUC1 CT aa residues 37–69. These are referred to as peptide #1, peptide #2, and peptide #3, respectively. Additionally, HSPTMDRSESYPPYT EYKHSGAV, a 23 aa peptide corresponding to the scrambled sequence of the overlapping regions of peptide #1, peptide #2, and peptide #3, and referred to as control peptide, and a 20 aa peptide containing one complete MUC1 TR (PDTRPAPGSTAPPAHGVTSA) and referred to as TR, were used. All peptides were synthesized, characterized, and purified to >95% purity by Genemed Synthesis, South San Francisco, CA, USA.

MUC1 CT peptide vaccinations and tumor challenge

Peptides were diluted in PBS, pH 7.4, at a concentration of 500 μg/ml. The mixture was vortexed vigorously for 5 min. Mice receiving B16.MUC1 tumors received one vaccination consisting of a combination of 50 μg of peptide #1, 50 μg of peptide #2, and 50 μg of peptide #3 in a total volume of 100 μl by s.c. injection between the scapulae. Control mice received one vaccination consisting of 50 μg of control peptide in a total volume of 100 μl by s.c. injection between the scapulae or no vaccination. Ten days following vaccination, 2×10^4 viable B16.MUC1 cells were injected subcutaneously between the scapulae. Mice receiving Panc02.MUC1 tumors received three vaccinations, at 7-day intervals, consisting of 50 μg of peptide #1 in a total volume of 100 μl, by s.c. injection between the scapulae. Control mice received three vaccinations, at 7-day intervals, consisting of 50 μg of control peptide in a total volume of 100 μl, by s.c. injection between the scapulae or no vaccination. Ten days following the third vaccination, 1×10^6 viable Panc02.MUC1 cells were injected s.c. between the scapulae. Tumor challenge results were evaluated as described in a previous section. It should be noted that the immunization protocols described above were selected after a series of preliminary investigations in which we evaluated immunization with different concentrations of peptides (control and experimental) up to 150 μg, with and without coadministration of GM-CSF as an adjuvant. The concentrations of peptides reported here gave optimal responses in preliminary studies with the different tumor types (B16 and Panc02), and surprisingly, there was no improvement in the observed immune responses by inclusion of GM-CSF (data not shown). Thus, the immunization protocols reported here did not use any adjuvant, and utilized different total amounts of peptides that were optimized for the different tumors evaluated here.

Immunohistochemical examination of tumors for MUC1 expression

For immunohistochemical evaluation of MUC1 expression on tumors obtained from vaccinated, control-vaccinated, and nonvaccinated animals, tumor tissue was fixed in a buffered formalin solution (100 ml formalin, 3.4 g NaH_2PO_4, and 10.3 g Na_2HPO_4)/1,000 ml, pH 7.3–7.4, embedded in paraffin and sectioned at a thickness of 5 μm. Tissue sections were assayed using a modification of previously described ABC immunohistochemical methods [29]. Briefly, tissue sections were deparaffinized in EZ-DeWax (Biogenex, San Ramon, CA, USA). Antigen unmasking was carried out by adding 10 mM citrate buffer and boiling for 10 min. The sections were then incubated in blocking serum for 20 min and primary mAb M2 (Sigma, St Louis, MO,

USA) was added and the samples were kept in a humid chamber at 4°C overnight. The slides were rinsed with PBS and incubated for 1 h with biotin-labeled secondary mAb. Endogenous peroxidase activity was blocked by incubating the samples in 3% H_2O_2 for 5 min. The slides were then incubated for 30 min at room temperature with ABC reagent (Vector Labs, Burlingame, CA, USA). The slides were then rinsed with PBS and incubated for 3–5 min with DAB substrate (Vector Labs), and observed closely for color to develop. Sections were then incubated for 10 min in 50 mM sodium bicarbonate, pH 9.6, followed by a 5–20 s incubation in DAB-enhancing solution (Vector Labs) and counterstained with Meyer's hematoxylin for 30 s. Cover slips were applied and the slides were examined under a Nikon E400 microscope (Nikon, Tokyo, Japan). Images were captured using a Nikon CoolPix 950 digital camera (Nikon).

Cytotoxic T lymphocyte assay

Detection of MUC1-specific CTLs was carried out by generation of effector cells from spleens and lymph nodes (axillary, brachial, inguinal, and mesenteric) harvested from vaccinated, control peptide–vaccinated, and nonvaccinated mice. Red blood cells were lysed using RBC lysis buffer (800 mg NH_4Cl, 80 mg EDTA, 80 mg $NaHCO_3$)/100 ml. EL4 and EL4.MUC1 target cells were labeled with 100 μCi $Na_2\ ^{51}CrO_4$ for 1.5 h and washed. Effector cells were incubated with 5×10^3 ^{51}Cr-labeled EL4 or EL4.MUC1 cells at various effector to target ratios (E/T) in triplicate in 96-well plates (Falcon) in a total volume of 200 µl. The plates were incubated for 6 h at 37°C and 5% CO_2. Following incubation, 50 µl culture supernatant was transferred to Lumaplates (Packard Instrument Company, Meriden, CT, USA). The Lumaplates were air dried overnight at room temperature, and radioactivity was measured using a TopCount NXT (Packard). Percentage specific lysis was determined by the following equation: percentage specific lysis = [(experimental ^{51}Cr release − spontaneous ^{51}Cr release) / (maximum ^{51}Cr release − spontaneous ^{51}Cr release)]×100%. "Experimental ^{51}Cr release" refers to ^{51}Cr release from targets mixed with effectors, "spontaneous ^{51}Cr release" refers to targets in medium only, and "maximum ^{51}Cr release" refers to targets lysed with 5% Triton X-100.

Statistical analysis

Tumor challenge studies were conducted two or more times. Survival data were pooled and the log-rank test was used for statistical analysis of survival. Additionally, the Cox regression analysis was used to confirm statistical differences among experimental groups and to verify that data from repeated experiments were statistically similar prior to being pooled. For CTL assays, statistical significance was determined by one-way ANOVA followed by Newman-Keuls multiple comparison test. For all statistical tests, a p value <0.05 was considered to be statistically significant.

Results

Surface expression of human MUC1 variants encoded by deletion constructs expressed in B16 and Panc02 cells

A schematic of recombinant proteins encoded by the MUC1 cDNA constructs used in this study is shown in Fig. 1. Surface expression of MUC1 isoforms by the

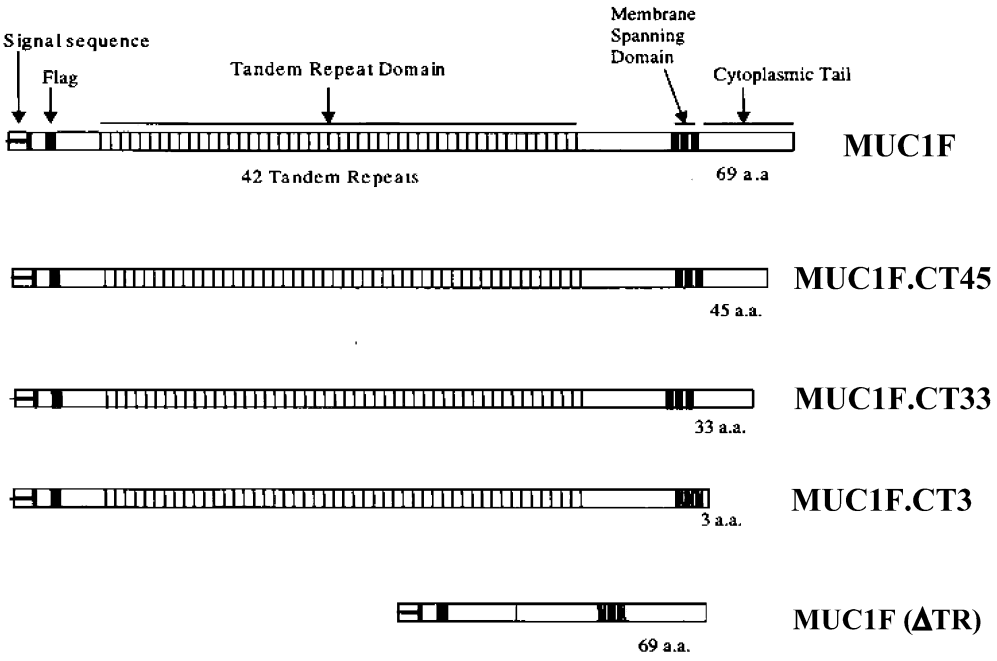

Fig. 1 Schematic representations of full-length MUC1, MUC1 cytoplasmic tail truncation constructs, and MUC1 tandem repeat deletion construct. Schematic diagram of MUC1 showing the locations of the Flag epitope tag, tandem repeat domain (TR), membrane-spanning domain, and cytoplasmic tail (CT). The full-length cytoplasmic tail of MUC1 is 69 aa in length. The cytoplasmic tail truncation constructs show deletions of the final 24 aa (MUC1.CT45), final 36 aa (MUC1.CT33), or the final 66 aa (MUC1.CT3). Also shown is the construct in which the TR of MUC1 has been removed (MUC1[ΔTR])

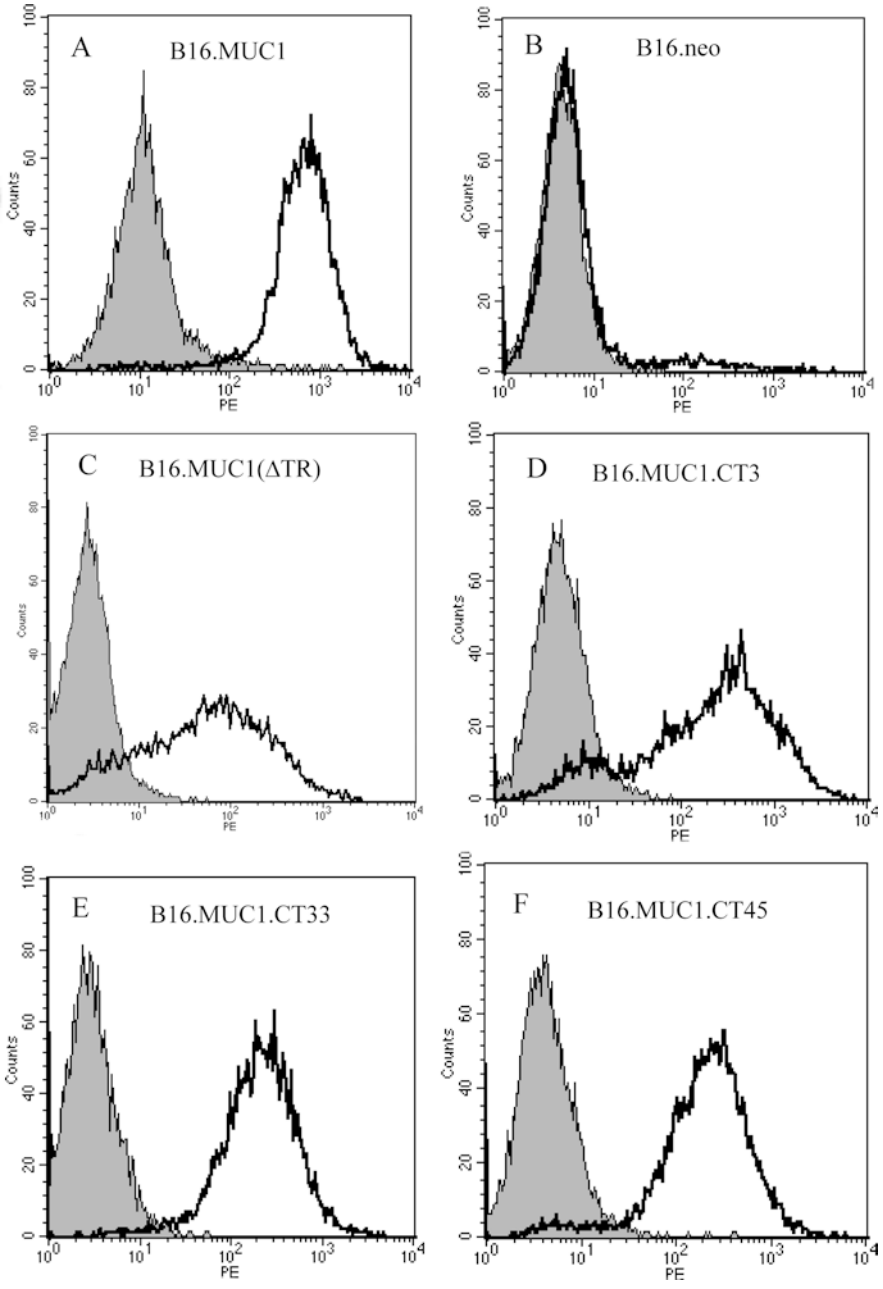

Fig. 2A–F Surface expression of MUC1 deletion constructs on B16 cells. Flow cytometry analyses of B16 cells transfected with MUC1 (**A**), neoexpressing vector only (**B**), MUC1.CT3 (**C**), MUC1.CT33 (**D**), MUC1.CT45 (**E**), or MUC1(ΔTR) (**F**) were conducted by incubating cells with mAb M2 and PE-conjugated secondary antibody (*solid line*) or secondary antibody only (*shaded peaks*). B16 cells expressing MUC1 constructs showed reactivity with M2 antibody while B16.neo did not

different cell lines was evaluated by flow cytometry (Figs. 2 and 3). Control transfected B16 (B16.neo) and Panc02 (Panc02.neo) were unreactive with the M2 or anti-MUC1 HMFG-2 antibodies (Figs. 2B and 3B). At least two representative clones of each cell line, with similar in vitro growth rates, and expressing similar levels of the MUC1 isoforms (as determined by western blot analysis) were selected for further studies.

Experimental system

Wt C57BL/6 mice produce MUC1-specific immune responses, when challenged with MUC1-expressing B16 tumor cells (Fig. 4A) [17, 18], and Panc02 tumor cells (Fig. 5A) [20, 23], which are lacking in MUC1.Tg mice because of immunological tolerance, as evidenced in vivo by differences in survival among these strains. The conclusion that immune responses in wt mice against B16.MUC1 and Panc02.MUC1 are MUC1-specific is supported by the finding that survival of wt and MUC1.Tg mice challenged with B16.neo or Panc02.neo cells were statistically indistinguishable ($p > 0.05$) (Figs. 4B and 5B). In studies presented here, we sought to identify immunodominant epitopes for MUC1-specific immune responses in vivo by challenging mice with tumors expressing recombinant isoforms of MUC1 that lacked defined regions of the CT

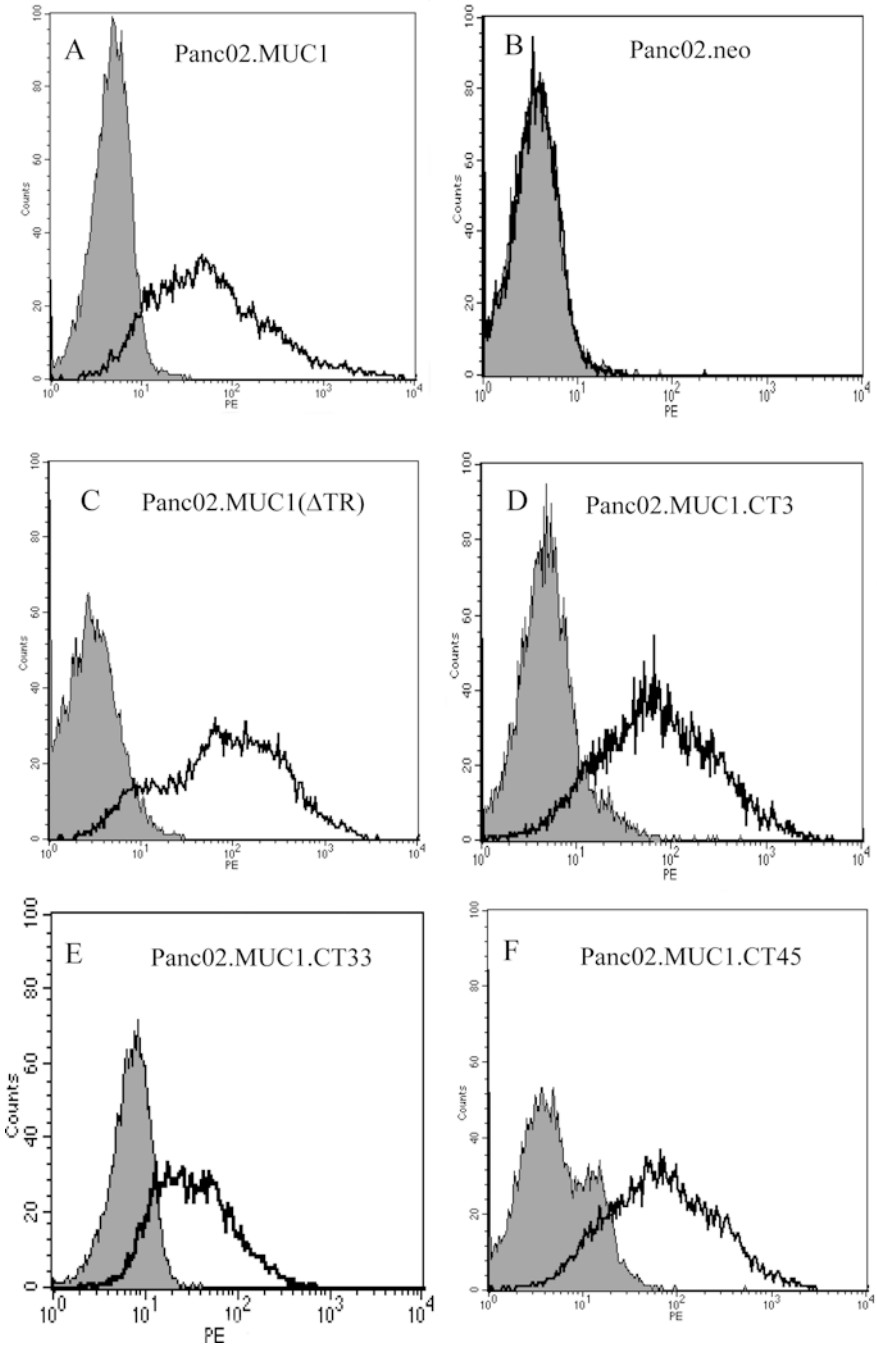

Fig. 3A–F Surface expression of MUC1 deletion constructs on Panc02 cells. Flow cytometry analyses of Panc02 cells transfected with MUC1 (**A**), neoexpressing vector only (**B**), MUC1.CT3 (**C**), MUC1.CT33 (**D**), MUC1.CT45 (**E**), or MUC1(ΔTR) (**F**) were conducted by incubating cells with mAb M2 and PE-conjugated secondary antibody (*solid line*) or secondary antibody only (*shaded peaks*). Panc02 cells expressing MUC1 constructs showed reactivity with M2 antibody while Panc02.neo did not

or TR. Evidence that the deleted portion of MUC1 contributed to MUC1-specific immune responses was obtained when survival curves for wt animals challenged with B16 or Panc02 tumor cells expressing the deleted forms were similar to those of MUC1.Tg mice or mice challenged with (MUC1-negative) B16.neo or Panc02.neo. If the deleted portion of MUC1 was not required for MUC1-specific rejection of tumors, then the survival curves more closely resembled those of wt mice challenged with B16 or Panc02 cells expressing full-length MUC1.

Contribution of the cytoplasmic tail of MUC1 to MUC1-specific immune responses against MUC1-expressing B16 tumors

Initially, we examined the CT for epitopes that contributed to $CD4^+$ MUC1-specific immune responses directed against B16.MUC1 tumor cells. Wt and MUC1.Tg mice were challenged s.c. with 2×10^4 B16 tumor cells expressing a construct in which all but 3 aa of the C-terminus were deleted (B16.MUC1.CT3). Data in Fig. 4C show that elimination of the CT, eliminated

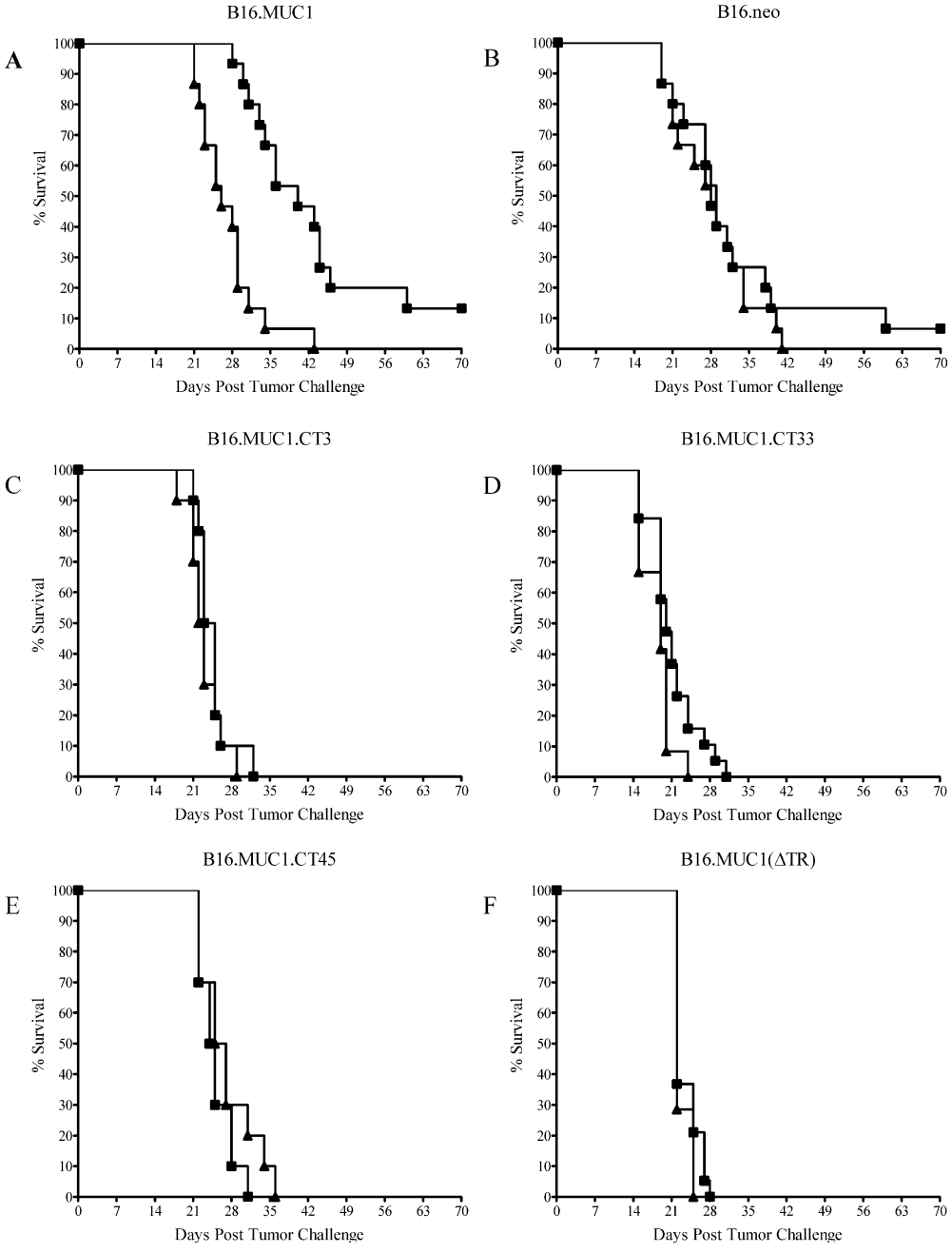

Fig. 4A–F The cytoplasmic tail and the tandem repeat of MUC1 are critical to MUC1-specific immune responses against the B16.MUC1 tumor cell line. (Wt mice represented by *solid squares* and MUC1.Tg mice by *solid triangles*, in all graphs.) **A** Wt ($n=15$) and MUC1.Tg ($n=15$) mice were challenged with 2×10^4 B16.MUC1 cells s.c. **B** Wt ($n=15$) and MUC1.Tg ($n=15$) mice were challenged with 2×10^4 B16.neo cells s.c. **C** Wt ($n=10$) and MUC1.Tg ($n=10$) mice were challenged with 2×10^4 B16.MUC1.CT3 cells s.c. **D** Wt ($n=18$) and MUC1.Tg ($n=12$) mice were challenged with 2×10^4 B16.MUC1.CT33 cells s.c. **E** Wt ($n=10$) and MUC1.Tg ($n=10$) mice were challenged with 2×10^4 B16.MUC1.CT45 cells s.c. **F** Wt ($n=19$) and MUC1.Tg ($n=15$) mice were challenged with 2×10^4 B16.MUC1(ΔTR) cells s.c. on day 0 and tumor growth was measured over time. Data were pooled from at least two independent experiments conducted at different times (after determining that there were no significant differences in data from the same experimental groups, by Cox regression analysis). **A** Statistically significant increase (log-rank) in survival was observed in wt mice compared with MUC1.Tg mice ($p<0.001$). **B–F** No statistically significant differences in survival were observed between wt and MUC1.Tg mice ($p>0.05$)

MUC1-specific immune responses toward B16.MUC1 ($p>0.05$), suggesting that the CT is critical to the immunological rejection of these tumors.

We next evaluated immune responses to two additional constructs expressed by B16 tumor cells in which the final 36 aa (MUC1.CT33) or the final 24 aa (MUC1.CT45) of the CT was deleted. As shown in Fig. 4(D, E), MUC1-specific tumor rejection was not detected in wt mice challenged with B16 tumor cells that expressed either MUC1.CT33 ($p>0.05$) or MUC1.CT45 ($p>0.05$). These results suggest that the final 24 aa segment of the CT contains an epitope or region that is critical for immunological responses that mediate rejection of MUC1-expressing B16 tumor cells.

Contribution of the tandem repeat region to MUC1-specific immune responses against the B16.MUC1 tumor cell line

The contribution of the TR to MUC1-specific immune responses directed against B16.MUC1 tumor cells was similarly evaluated. Wt and MUC1.Tg mice were challenged s.c. with 2×10^4 B16 cells expressing a construct in which the large, extracellular TR portion was deleted (B16.MUC1[ΔTR]). Wt mice challenged with B16.MUC1(ΔTR) experienced survival similar to MUC1.Tg mice challenged with this tumor cell line ($p>0.05$) (Fig. 4F). These findings confirm that the TR is an important immunological target during rejection of

B16.MUC1 tumor cells, when expressed together with the final 24 aa segment of the CT (Fig. 4A).

Contribution of the cytoplasmic tail to MUC1-specific immune responses against the Panc02.MUC1 tumor cell line

In parallel studies, we evaluated the contribution of the CT to MUC1-specific immune responses against MUC1-expressing Panc02 tumor cells. Previous studies demonstrated that wt C57BL/6 mice produce $CD8^+$ T cell–mediated, MUC1-specific immune responses that reject MUC1-expressing Panc02 tumors, whereas MUC1.Tg mice do not reject these tumors (Fig. 5A) [20, 23]. Groups of wt and MUC1.Tg mice were challenged s.c. with 1×10^6 Panc02 tumor cells expressing MUC1.CT3. Consistent with the results of studies utilizing B16.MUC1.CT3 tumor cells, the data in Fig. 5C show that elimination of the CT eliminated MUC1-specific immune responses toward Panc02.MUC1 ($p > 0.05$).

To further delineate the contribution of the CT to MUC1-specific immune responses against MUC1-expressing Panc02 tumor cells, we evaluated the survival of wt and MUC1.Tg mice following challenge with 1×10^6 Panc02 tumor cells expressing MUC1.CT33. In contrast to results with B16 tumor cells expressing this form of MUC1, wt mice rejected these tumors (Fig. 5D) and MUC1.Tg mice did not reject these tumors ($p < 0.05$).

We next examined the survival of wt and MUC1.Tg mice following challenge with 1×10^6 Panc02 tumor cells expressing MUC1.CT45. The wt mice rejected MUC1.CT45-expressing tumors, in a MUC1-specific manner (Fig. 5E), whereas MUC1.Tg mice did not ($p < 0.05$). These findings suggest an epitope or region between aa 3 and 33 of the CT contributes to immunological responses necessary to reject MUC1-expressing Panc02 tumors.

Contribution of the tandem repeat region to MUC1-specific immune responses against the Panc02.MUC1 tumor cell line

We evaluated the contribution of the TR to MUC1-specific immune responses to Panc02.MUC1. Wt and MUC1.Tg mice were challenged with 1×10^6 Panc02 tumor cells expressing MUC1(ΔTR). Prolonged survival was observed for wt mice compared with MUC1.Tg mice, following challenge with Panc02.MUC1(ΔTR) tumor cells (Fig. 5F) ($p < 0.05$). These results indicate that the TR portion of MUC1 is not required to reject Panc02.MUC1 tumor cells in a MUC1-specific manner.

Epitope prediction

Studies presented herein demonstrate that epitopes in the MUC1 CT are critical for the immunological rejection of two distinct MUC1-expressing tumors. We evaluated the MUC1 CT aa sequence using two independent Web-based algorithms that predict potential binding to MHC class I and class II molecules. The algorithms and Web-based programs for analyzing potential binding to MHC molecules are found at http://www.uni-tuebingen.de/uni/kxi/ [27] and http://bimas.dcrt.nih.gov/molbio/hla_bind/ [28]. Epitope prediction was conducted for the following 72 aa sequence of the MUC1 CT and adjoining sequence: CQCRRKNYGQLDIFPARDTYHPMSEYPTYHT-HGRYVPPSSTDRSPYEKVSAGNGGSSLSYTNPA-VAAASANL. Multiple peptides within the regions required for rejection of Panc02.MUC1 (aa 3–33) and B16.MUC1 (aa 45–69) were predicted to exhibit high binding affinity for murine MHC class I. The results for binding to K^b and D^b are summarized in Table 1. Similar analyses revealed numerous putative epitopes within these regions predicted to bind human HLA molecules (Table 2). These results demonstrate that the CT of MUC1, which has a high degree of identity between humans and mice, may be utilized for future immunotherapy protocols that target MUC1-positive tumors in humans.

MUC1-specific immune response elicited against B16.MUC1 in MUC1.Tg mice following vaccination with MUC1 CT peptides

To evaluate efficacy of CT peptide vaccination, MUC1.Tg mice were challenged with a lethal dose of MUC1-expressing B16 tumor cells 10 days following vaccination. As shown in Fig. 6A, MUC1.Tg mice vaccinated with MUC1 CT peptides demonstrated a significant increase in survival compared with MUC1.Tg mice vaccinated with control peptide or nonvaccinated mice ($p < 0.05$). Organs expressing endogenous MUC1 were examined at necropsy and histologically to determine whether the vaccination also elicited detectable autoimmunity. No evidence of autoimmunity was detected in any of the animals.

MUC1 expression in tumors was evaluated among vaccinated, control peptide–vaccinated, and nonvaccinated mice. Serial sections of tumor tissue were evaluated for expression of MUC1 immunohistochemically. Tumors from control peptide–vaccinated MUC1.Tg mice challenged with B16.MUC1 showed significantly greater expression of MUC1 than vaccinated mice challenged with B16.MUC1 (Fig. 7A, C). Similar results were observed in tumors obtained from nonvaccinated mice challenged with B16.MUC1 compared with vaccinated mice challenged with B16.MUC1 (data not shown).

MUC1-specific immune response elicited to Panc02.-MUC1 in MUC1.Tg mice following vaccination with peptide derived from MUC1 CT sequence

The efficacy of MUC1 CT peptide vaccinations to protect against MUC1-expressing Panc02 tumors was

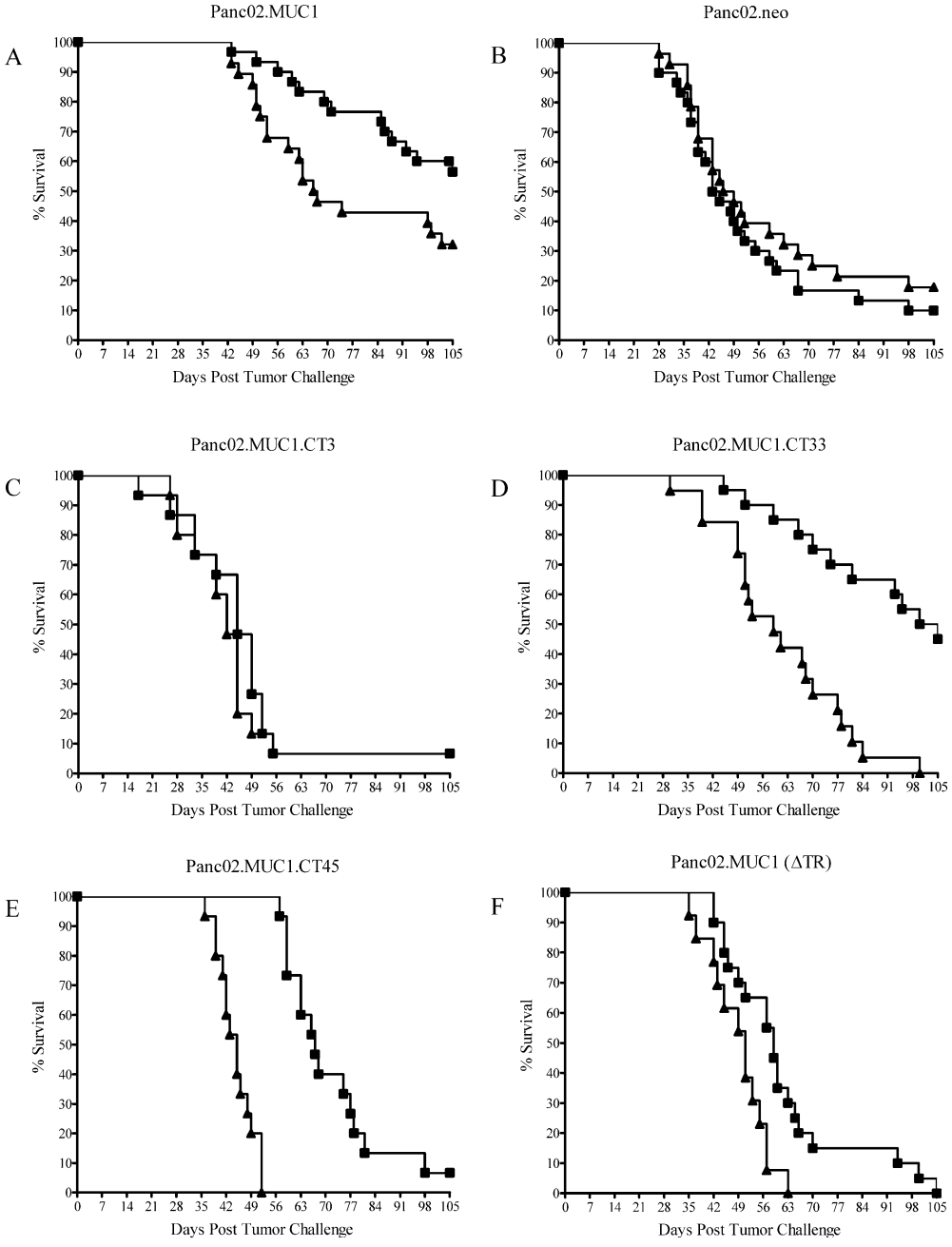

Fig. 5A–F The cytoplasmic tail, but not the tandem repeat, of MUC1 is critical to MUC1-specific immune responses against the Panc02.MUC1 tumor cell line. (Wt mice represented by *solid squares* and MUC1.Tg mice by *solid triangles*, in all graphs.) **A** Wt ($n=20$) and MUC1.Tg ($n=20$) mice were challenged with 1×10^6 Panc02.MUC1 cells s.c. **B** Wt ($n=20$) and MUC1.Tg ($n=19$) mice were challenged with 1×10^6 Panc02.neo cells s.c. **C** Wt ($n=15$) and MUC1.Tg ($n=15$) mice were challenged with 1×10^6 Panc02.MUC1.CT3 cells s.c. **D** Wt ($n=20$) and MUC1.Tg ($n=19$) mice were challenged with 1×10^6 Panc02.MUC1.CT33 cells s.c. **E** Wt ($n=15$) and MUC1.Tg ($n=15$) mice were challenged with 1×10^6 Panc02.MUC1.CT45 cells s.c. **F** Wt ($n=20$) and MUC1.Tg ($n=13$) mice were challenged with 1×10^6 Panc02.MUC1(ΔTR) cells s.c. on day 0 and tumor growth was measured over time. Data were pooled from at least two independent experiments conducted at different times (after determining that there were no significant differences in data from the same experimental groups, by Cox regression analysis). **A, D–F** Statistically significant increases (log-rank) in survival were observed in wt mice compared with MUC1.Tg mice (**A**, $p=0.01$; **D–E**, $p<0.0001$; **F**, $p=0.0058$). **B** and **C** No statistically significant differences in survival were observed between wt and MUC1.Tg mice ($p>0.05$)

evaluated by challenging MUC1.Tg mice with a lethal dose of Panc02.MUC1 tumor cells 10 days following the final vaccination. As shown in Fig. 6B, MUC1.Tg mice vaccinated with MUC1 CT peptide #1 demonstrated a significant increase in survival compared with MUC1.Tg mice vaccinated with control peptide or nonvaccinated mice ($p<0.05$). Organs from these mice examined at necropsy and histologically showed no evidence of autoimmunity (data not shown).

Serial sections of tumor tissue derived from vaccinated, control peptide–vaccinated, and nonvaccinated mice were evaluated by immunohistochemistry for expression of MUC1. Tumors from control peptide–vaccinated MUC1.Tg mice challenged with Panc02.-MUC1 showed significantly more MUC1 expression than vaccinated mice challenged with Panc02.MUC1 (Fig. 7E, G). Similar reactivity was observed in tumors obtained from nonvaccinated mice challenged with Panc02.MUC1 compared with vaccinated mice challenged with Panc02.MUC1 (data not shown). These results, combined with survival data, suggest that MUC1-expressing tumor cells were eliminated, and that tumor growth was primarily due to MUC1-negative variants. Moreover, the observed anti-MUC1 immune responses were apparently tumor-specific, as there was no evidence of autoimmune reactions.

Table 1 Candidate murine epitopes derived from the MUC1 CT aa sequence

Predicted epitope	Position of first residue	H-2 molecule	SYFPEITHI score[1]	BIMAS score[2]
CRRKNYGQL	3	H2-D^b	20	10.171
VSAGNGGSS	49	H2-D^b	14	No score obtained
LSYTNPAVA	58	H2-D^b	13	33.480
SAGNGGSSL	50	H2-D^b	18	22.176
AVAAASANL	64	H2-D^b	12	10.088
VSAGNGGSSL	49	H2-D^b	25	Decamer option not available
PAVAAASANL	63	H2-D^b	15	Decamer option not available
LSYTNPAVAA	58	H2-D^b	14	Decamer option not available
FPARDTYHPM	14	H2-D^b	13	Decamer option not available
RRKNYGQL	4	H2-K^b	22	Octamer option not available
DRSPYEKV	42	H2-K^b	18	Octamer option not available
ARDTYHPM	16	H2-K^b	17	Octamer option not available
SSLSYTNP	56	H2-K^b	14	Octamer option not available
CRRKNYGQL	3	H2-K^b	No score obtained	1.440
AVAAASANL	64	H2-K^b	No score obtained	1.210
SAGNGGSSL	50	H2-K^b	No score obtained	1.100
SEYPTYHTH	24	H2-K^b	No score obtained	1.100

[1]Score obtained using program developed by Rammensee et al. [27]
[2]Score obtained using program developed by Parker et al. [28]. Score for H2-D^b obtained using D^b revised

MUC1.Tg mice develop MUC1-specific cell-mediated but not humoral immune responses following vaccination with MUC1 CT peptides

To investigate whether observed anti-MUC1 immune responses were mediated by humoral and/or cell-mediated mechanisms, ELISA and CTL assays were performed. For ELISA, sera from vaccinated, control peptide–vaccinated, and nonvaccinated mice challenged with MUC1-expressing B16 or Panc02 tumors were tested for production of antibodies to peptide #1, peptide #2, peptide #3, control peptide, and TR peptide. There were no statistically significant differences in antibody responses to peptides among the vaccinated, control peptide–vaccinated, and nonvaccinated mice ($p > 0.05$), with one exception. Following challenge with MUC1-expressing Panc02 tumor cells, nonvaccinated mice produced significantly more peptide-specific antibodies to peptides compared with vaccinated or control peptide–vaccinated mice ($p < 0.05$) (data not shown).

To evaluate generation of MUC1-specific CTL, spleens and lymph nodes were harvested from vaccinated, control peptide–vaccinated, and nonvaccinated MUC1.Tg mice challenged with MUC1-expressing B16 or Panc02 tumors. MUC1-specific cytolytic activity was measured against EL4 or EL4.MUC1 cells. CTL were restimulated in vitro for 10 days with EL4 or EL4.MUC1 cells prior to analysis. Cytotoxic responses to EL4.MUC1 cells were similarly low among control peptide–vaccinated and nonvaccinated mice challenged with B16.MUC1 or Panc02.MUC1 (Fig. 8A; $p > 0.05$). In contrast, lymphocytes obtained from vaccinated mice challenged with B16.MUC1 or Panc02.MUC1 demonstrated significant lysis of EL4.MUC1 cells (Fig. 8A; $p < 0.05$). These responses were specific for MUC1 because there were no differences of EL4 cell lytic activity among vaccinated, control peptide–vaccinated, and nonvaccinated mice challenged with MUC1-expressing B16 or Panc02 tumor cells (Fig. 8B; $p > 0.05$).

Discussion

Previous studies of MUC1-based cancer vaccines [10, 11, 13] were focused on the TR domain, as in vitro assays determined that the TR region contained immunodominant epitopes for MUC1-specific antibodies and CTLs. Recent studies demonstrated epitopes of MUC1 outside of the TR region that may be potentially important immunogenic epitopes of this tumor-associated antigen including one epitope in the signal sequence and a second in the degenerate TR region described by Brossart et al. [14, 15], and three epitopes outside of the TR described by Heukamp et al. [16]. In results presented here, we show that different portions of the CT of MUC1 are critical to the production of cellular responses that mediate immunological rejection of MUC1-expressing B16 melanoma and Panc02 pancreatic carcinoma tumor cells. We also demonstrate that vaccination of immunologically tolerant MUC1.Tg mice with peptides derived from the aa sequence of the MUC1 CT elicits a protective immune response that significantly prolongs survival of these mice following challenge with MUC1-expressing B16 or Panc02 tumor cells.

Initial studies evaluated in vivo immune responses against B16 and Panc02 tumor cell lines that expressed

Table 2 Candidate human epitopes for MUC1 CT aa sequence

Predicted epitope	Position of first residue	HLA molecule	SYFPEITHI sore[1,3]	BIMAS score[2,3]
SLSYTNPAV	57	HLA-A*0201	23	69.552
SAGNGGSSL	50	HLA-A*0201	20	0.297
YTNPAVAAA	60	HLA-A*0201	20	0.730
AVAAASANL	64	HLA-A*0201	20	1.869
SYTNPAVAAA	59	HLA-A*0203	27	HLA type not available
DTYHPMSEY	18	HLA-A1	22	1.250
GNGGSSLSY	52	HLA-A1	22	0.625
PTYHTHGRY	27	HLA-A1	21	0.125
DIFPARDTY	12	HLA-A1	20	5.000
PSSTDRSPY	38	HLA-A1	20	0.075
AGNGGSSLSY	51	HLA-A1	23	No score obtained
STDRSPYEKV	40	HLA-A1	20	No score obtained
DTYHPMSEY	18	HLA-A26	32	HLA type not available
DIFPARDTY	12	HLA-A26	30	HLA type not available
DIFPARDTYH	12	HLA-A26	20	HLA type not available
DIFPARDTY	12	HLA-A3	22	0.900
KVSAGNGGSS	48	HLA-A3	21	No score obtained
YVPPSSTDR	35	HLA-A68.1	HLA type not available	300.000
RRKNYGQL	4	HLA-B*08	20	HLA type not available
SPYEKVSA	44	HLA-B*08	20	HLA type not available
SPYEKVSAG	44	HLA-B*08	22	HLA type not available
CRRKNYGQL	3	HLA-B*2705	23	HLA type not available
GRYVPPSST	30	HLA-B*2705	18	1000.000
ARDTYHPMS	13	HLA-B*2705	12	200.000
KNYGQLDIF	6	HLA-B*2705	17	150.000
GQLDIFPAR	9	HLA-B*2705	17	100.000
RRKNYGQLD	4	HLA-B*2705	14	60.000
CRRKNYGQL	3	HLA-B*2709	21	HLA type not available
HPMSEYPTY	21	HLA-B*3501	HLA type not available	60.000
SAGNGGSSL	50	HLA-B*5102	HLA type not available	60.500
RKNYGQLDIFPARDT	5	HLA-DRB1*0101	26	HLA type not available
RSPYEKVSAGNGGSS	43	HLA-DRB1*0101	26	HLA type not available
LDIFPARDTYHPMSE	11	HLA-DRB1*0101	24	HLA type not available
YEKVSAGNGGSSLSY	46	HLA-DRB1*0101	24	HLA type not available
GRYVPPSSTDRSPYE	33	HLA-DRB1*0101	22	HLA type not available
GSSLSYTNPAVAAAS	55	HLA-DRB1*0101	22	HLA type not available
YGQLDIFPARDTYHP	8	HLA-DRB1*0101	20	HLA type not available
RKNYGQLDIFPARDT	5	HLA-DRB1*0401	22	HLA type not available
YGQLDIFPARDTYHP	8	HLA-DRB1*0401	20	HLA type not available
YHPMSEYPTYHTHGR	20	HLA-DRB1*0401	20	HLA type not available
GRYVPPSSTDRSPYE	33	HLA-DRB1*0401	20	HLA type not available
YPTYHTHGRYVPPSS	26	HLA-DRB1*1101	25	HLA type not available
RKNYGQLDIFPARDT	5	HLA-DRB1*1101	22	HLA type not available
RSPYEKVSAGNGGSS	43	HLA-DRB1*1101	22	HLA type not available

[1]Score obtained using program developed by Rammensee et al. [27]
[2]Score obtained using program developed by Parker et al. [28]
[3]Minimum scores of 20 on the SYFPEITHI site or 60 on the BIMAS site were used. If a score met the requirement of one site, the score of the other site is listed even if the minimum score for that site was not achieved

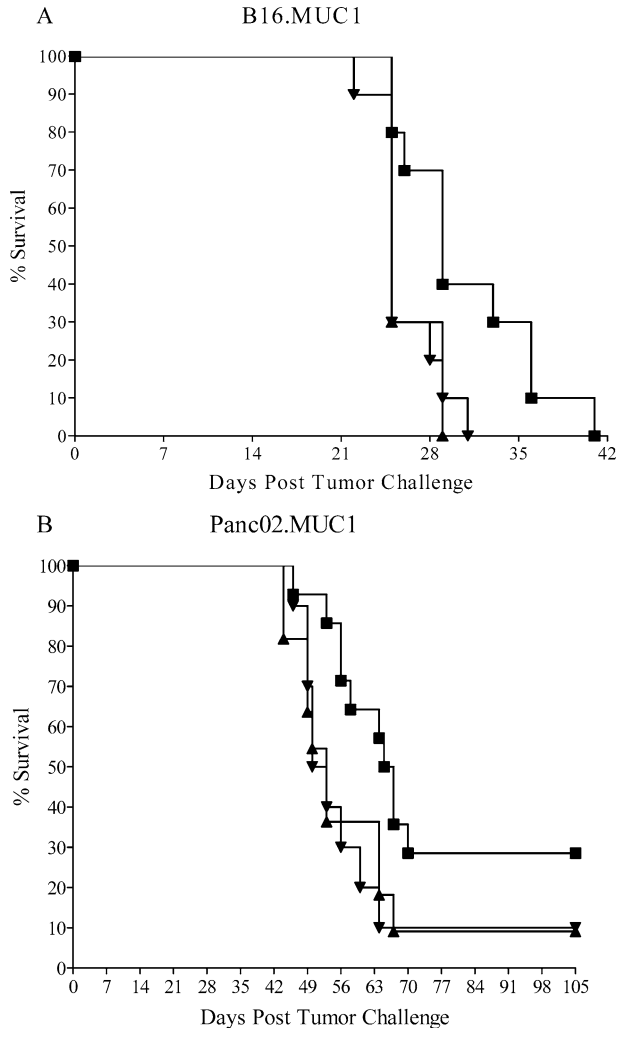

Fig. 6A, B Survival of vaccinated, control peptide–vaccinated, or nonvaccinated MUC1.Tg mice following challenge with B16.MUC1 or Panc02.MUC1 tumor cell lines. **A** MUC1.Tg mice received one vaccination with a combination of 50 μg peptide #1, 50 μg peptide #2, and 50 μg peptide #3 (*solid squares*; $n = 10$), or 50 μg control peptide (*solid pyramids*; $n = 10$), or were nonvaccinated (*solid inverted triangles*; $n = 10$). Ten days following vaccination, mice were challenged with 2×10^4 B16.MUC1 cells s.c. on day 0 and tumor growth was measured over time. **B** In parallel studies, MUC1.Tg mice received three vaccinations at 7 day intervals with 50 μg peptide #1 (*solid squares*; $n = 14$), or 50 μg control peptide (*solid pyramids*; $n = 11$), or were nonvaccinated (*solid inverted triangles*; $n = 10$). Ten days following the final vaccination, mice were challenged with 1×10^6 Panc02.MUC1 cells s.c. on day 0 and tumor growth was measured over time. Data were pooled from at least two independent experiments conducted at different times (after determining that there were no significant differences in data from the same experimental groups, by Cox regression analysis). A statistically significant increase in survival was observed in vaccinated mice compared with control-vaccinated or nonvaccinated mice (**A** and **B**; $p < 0.05$) as determined by log-rank statistics. No statistically significant differences of survival were observed between control-vaccinated and nonvaccinated MUC1.Tg mice (**A** and **B**; $p > 0.05$) as determined by log-rank statistics

MUC1 cDNA constructs with truncated CT or the deletion of the TR. The truncated CT constructs included MUC1.CT3, in which all but 3 aa were deleted from the CT; MUC1.CT33, which contained 33 aa in the CT; and MUC1.CT45, which contained 45 aa in the CT. Results of these studies (Figs. 4 and 5) demonstrated that the CT is critical for rejection of B16 and Panc02 tumors expressing MUC1; however, different portions of the CT were required to produce responses against these two tumor cell lines. Minimally, epitopes within 24 aa of the C-terminus and epitopes in the TR were required for immune responses that eliminated MUC1-expressing B16. In contrast, epitopes from a region that is between aa 3–33 on the C-terminal side of the transmembrane domain on the CT were required for generation of immune responses to MUC1-expressing Panc02 cells.

Algorithmic analyses of the MUC1 CT peptide sequence showed (Table 1) several putative epitopes associated with murine MHC I that could be presented to $CD8^+$ cells [27, 28]. Therefore, rejection of Panc02.-MUC1 may be directed against epitope(s) contained within aa 3–33 of the CT that are presented to $CD8^+$ cells by MHC I. It is notable that a number of peptides from the CT domain are likely to bind human MHC molecules (Table 2), a fact that supports further development of human tumor vaccines that target the CT.

The results of our in vivo studies using deletion constructs of MUC1 and algorithmic analyses led us to develop vaccine formulations using synthetic peptides based on the CT portion of MUC1. We evaluated different MUC1 CT peptide vaccine formulations for the two tumor cell lines used in these studies (Fig. 6, and data not shown). For the rejection of B16.MUC1 tumors, a single vaccination with a combination of peptides #1, #2, and #3 was shown to be effective. Enhanced rejection of Panc02.MUC1 tumors required a series of three vaccinations with peptide #3. These results provide further insight into earlier observations that different immunological pathways and effector cells lead to rejection of these two tumor cell lines in vivo [17, 20], by demonstrating that different epitopes within the MUC1 protein were required to generate effective immune responses against these two tumor cell types. Surprisingly, addition of GM-CSF to vaccine formulations did not enhance efficacy (data not shown), which suggests that these 20-residue peptides are immunogenic in the absence of adjuvants. As such, the vaccination conditions reported here represent the minimal requirements for producing a tumor-specific anti-MUC1 response in the context of the immunologically tolerant MUC1.Tg mouse. It will be important to determine in future studies if other adjuvants or extended immunization protocols improve the efficacy of these peptides as vaccine components.

Immunohistochemical studies (Fig. 7) of tumors that arose late in vaccinated mice showed that these tumors expressed low or undetectable levels of MUC1, whereas B16.MUC1 or Panc02.MUC1 tumors arising in control peptide–vaccinated or nonvaccinated mice were positive for MUC1. Furthermore, the degree of autoimmunity was determined independently by a pathologist who histologically evaluated MUC1-positive and other

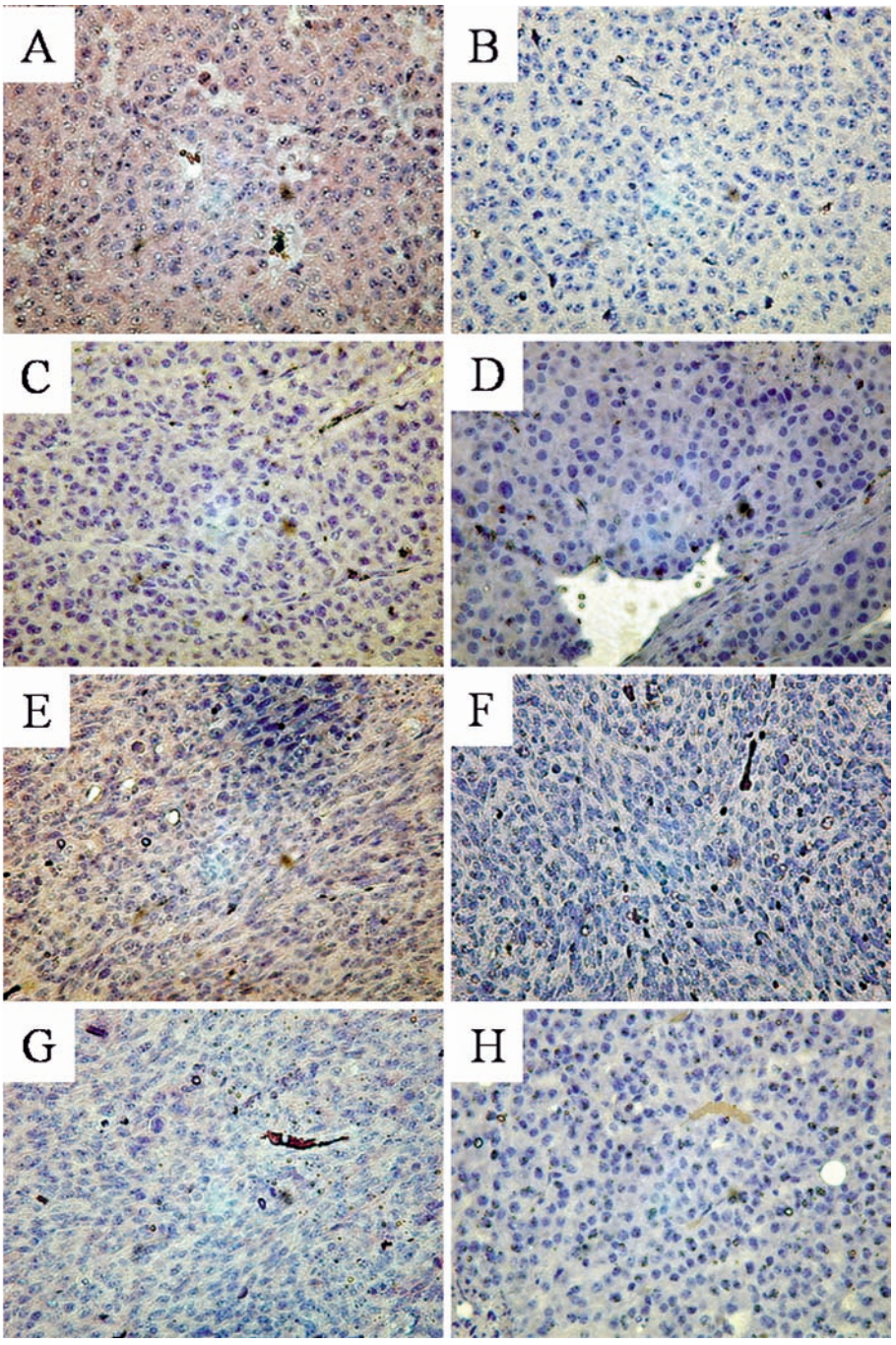

Fig. 7A–H Immunohistochemical staining of B16.MUC1 and Panc02.MUC1 tumors from vaccinated and nonvaccinated MUC1.Tg mice. Serial sections of tumor tissue were stained for expression of MUC1 with M2 antibody. Tumors from control-vaccinated MUC1.Tg mice challenged with B16.MUC1 (**A**) or Panc02.MUC1 (**E**) were significantly more MUC1-positive than vaccinated mice challenged with B16.MUC1 (**C**) or Panc02.MUC1 (**G**). Similar reactivity was observed between tumors obtained from nonvaccinated mice challenged with B16.MUC1 or Panc02.MUC1 compared with vaccinated mice challenged with B16.MUC1 or Panc02.MUC1 (data not shown). Specificity of the antibody staining was confirmed by the lack of staining in tumors from either control stained group (**B**, **D**, **F**, and **H**)

organs of vaccinated and nonvaccinated mice for leukocyte infiltration, inflammation, or other evidence of autoimmunity. No differences in leukocyte infiltration or other evidence of autoimmunity were observed between vaccinated and nonvaccinated animals. These findings suggest that the observed anti-MUC1 immune responses are tumor-specific.

To determine whether the tumor immune responses were antibody- or T cell–mediated, ELISA and CTL assays were conducted on sera and lymphocytes collected from the various groups of vaccinated and nonvaccinated mice. Results of CTL assays demonstrated detectable tumor-specific anti-MUC1 immune responses (Fig. 8). ELISA showed no statistically significant differences of antibody responses to peptides among the vaccinated, control-vaccinated, and nonvaccinated mice, with one notable exception. Following tumor challenge, nonvaccinated mice challenged with Panc02.MUC1 generated significantly higher levels of anti-MUC1 antibodies than vaccinated mice. These findings further support and extend previous findings in our laboratory suggesting that effective MUC1-specific immune responses directed against B16.MUC1 or Panc02.MUC1 are T cell–mediated and that the development of a

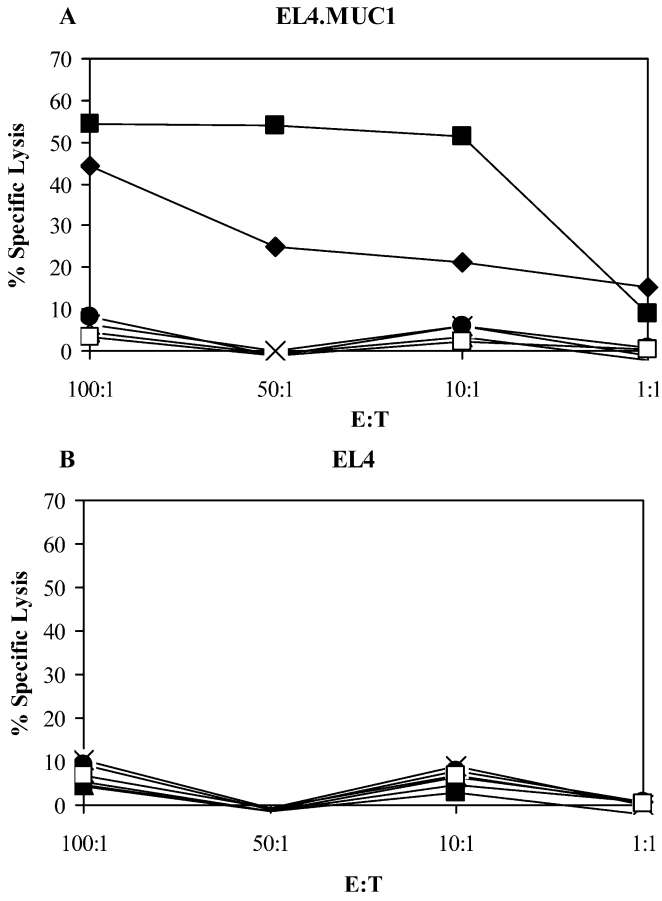

Fig. 8A, B MUC1-specific CTL activity. Spleens and lymph nodes were harvested from vaccinated, control peptide–vaccinated, and nonvaccinated MUC1.Tg mice challenged with B16.MUC1 or Panc02.MUC1 tumor cells and were analyzed for generation of MUC1-specific CTL by ^{51}Cr-release assays. **A** Vaccinated mice challenged with B16.MUC1 (*solid squares*) or Panc02.MUC1 (*solid diamonds*) generated significantly higher percentage specific lysis of EL4.MUC1 cells than control peptide–vaccinated mice challenged with B16.MUC1 (*solid triangles*), or Panc02.MUC1 (*solid circles*), or nonvaccinated mice challenged with B16.MUC1 (×) or Panc02.MUC1 (*open squares*); $p < 0.05$. **B** There was no statistically significant difference in lysis of EL4 cells among cells obtained from vaccinated, control peptide–vaccinated, and nonvaccinated mice; $p > 0.05$. Statistical significance ($p < 0.05$) was determined by one-way ANOVA followed by Newman-Keuls multiple comparison test

humoral anti-MUC1 immune response may be detrimental to tumor immunity in this experimental system [17, 20–23].

Based on the observation that wt mice do not reject B16.MUC1(ΔTR) tumors, we hypothesized that the TR region of MUC1 was also required for the immunological rejection of B16.MUC1 tumors. However, vaccination of wt or MUC1.Tg mice with MUC1 TR peptide alone did not affect survival following challenge with B16.MUC1 (data not shown). These peptides may lack important post-translational modifications (i.e., glycosylation) or proper tertiary structure necessary for immunogenicity. Alternatively, production of predominantly CD4$^+$ T cell–mediated responses that are required for rejection of B16.MUC1 (discussed further below) may require multiple immunogenic epitopes, some of which may serve as helper epitopes. These results may provide insight into why previous immunization attempts utilizing antigen derived from solely from the TR domain of MUC1 were ineffective in producing protective immunity to tumors in humans.

Differential antigen processing and expression of MHC class I and II may explain the distinct immune responses and recognition of different epitopes on MUC1 expressed by the syngeneic melanoma (B16) and pancreatic carcinoma (Panc02). Previous characterization of the in vivo cellular responses that mediate rejection of B16.MUC1 and Panc02.MUC1 led us to conclude that MUC1 epitopes recognized in rejection of B16.MUC1 are presented to CD4$^+$ cells via MHC II, and MUC1 epitopes recognized during rejection of Panc02.MUC1 are presented to CD8$^+$ cells via MHC I [21, 22].

There are no programs in the public domain that allow prediction of peptides that bind to murine MHC class II; however, as is shown in Table 2, there are numerous peptides in this region that are likely to bind to human MHC class II. Because of the wide range of peptide lengths and sequences that can bind to MHC class II molecules, it is likely that some peptides in this region also bind to I-A^b. B16 cells express modest levels of MHC class I and class II in the presence of IFN-γ; however, Panc02 cells only express MHC class I and do not express detectable class II [21, 22]. One implication of these results is that in cases of expression of both MHC I and II by tumor cells, CD4$^+$ T cell–mediated responses directed toward MHC II–associated epitopes may predominate. In the absence of MHC class II expression, mice develop CD8$^+$ T cell–mediated responses against MHC class I–associated epitopes.

Another potential interpretation of our results that cannot be ruled out at this point is that rejection of B16.MUC1 by CD4$^+$ cells may be directed against epitopes within the TR, and deletions of the CT domain may cause alterations in the post-translational processing of the TR, thereby altering processing and presentation of epitopes from this region of MUC1. One construct of MUC1 that was utilized in these studies had the TR portion removed (MUC1[ΔTR]). The TR portion of MUC1 comprises a 20-aa sequence (PDTRPAPGSTAPPAHGVTSA) that repeats 42 times, and possesses a large number of potential *O*-glycosylation sites [30–32]. Cytoplasmic motifs have been reported to direct intracellular trafficking and subcellular localization of some proteins [33, 34]. The aa sequence of the MUC1 CT is highly conserved across species [35] and is hypothesized to play a role in the post-translational processing, subcellular localization, signal transduction, and intracellular localization of MUC1. Thus, alterations to the CT may affect some aspects of trafficking of MUC1 through the Golgi and thereby influence glycosylation of the TR domain. Pemberton et al. [24] showed that the CT is not required for apical

expression of the molecule, which suggests that general trafficking of MUC1 from the endoplasmic reticulum to the surface is not dependent on the TR. Similarly, we determined by flow cytometric analysis that all four isoforms of MUC1 containing truncations of the CT and the TR deletion were expressed at the cell surface. This suggests that the intracellular trafficking of these proteins from the endoplasmic reticulum to the cell surface was not disrupted by these alterations. Another possibility is that the CT may influence the process by which surface-associated MUC1 is internalized and processed for antigen presentation by MHC class I or class II. These possibilities will be addressed in future studies.

Other factors that involve the CT, such as phosphorylation and associated signal transduction events, may be important to the biological function of MUC1 and may contribute to differences between MUC1 expressed by tumors and normal cells [24, 36–40]. Beta catenin has been shown to bind to a consensus site found in the CT; GSK-3β can phosphorylate a serine residue in the CT; and there is a consensus Grb2/Sos binding site (including a potentially phosphorylated tyrosine residue) in the CT [38–40]. Tumor-associated modifications of these sites may affect their immunogenicity, ability to associate with class I or class II molecules, or availability for processing and presentation at the cell surface in the context of MHC molecules. Conversely, deletion of portions of the CT might also affect these events and also influence antigen processing and presentation. Thus, the biochemical nature of epitopes contained in the CT of tumor-associated MUC1 should be fully defined in future studies.

Our findings are important to the rational design of antitumor immunotherapies. Most preclinical studies that evaluate immune responses to tumor antigens characterize only one tumor type. Many tumor-associated antigens are similar to MUC1 in that they are expressed by different types of malignancies. If the immunological requirements, cellular and molecular, for rejecting different types of tumors are distinct, then different vaccine strategies will be required to produce effective antitumor immune responses. The further development of effective antitumor immunotherapies requires greater understanding of the epitopes that are antigenic and the molecular and cellular components necessary for the development of the proper immune response to reject the tumor that is targeted by the vaccine.

In summary, we have identified regions of MUC1 that are necessary for immunological rejection of two distinct syngeneic MUC1-expressing tumor cell lines utilizing in vivo assays. Based on those findings, we developed vaccination formulations using peptides derived from sequences of MUC1 that were determined to play an important role in immunological rejection of B16.MUC1 and Panc02.MUC1 tumors. The vaccinations significantly prolonged survival of MUC1.Tg mice following challenge with a lethal dose of B16.MUC1 or Panc02.MUC1. The observed immune responses were specific to tumor-associated forms of MUC1 because no evidence of autoimmunity was detected in any of the mice. Our results from these studies should prove to be useful for the development of effective, tumor-specific, anti-MUC1 immunotherapies.

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